Laptop Price Predictor: Minor Project Report

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Abstract

The Laptop Price Predictor is a machine learning application designed to estimate the market price of laptops based on various hardware specifications. This project leverages Linear Regression to identify the relationships between features like RAM, storage, processor speed, and the final price. The system provides a practical demonstration of applying regression models to real-world pricing scenarios, offering insights into how different specifications influence laptop costs. It includes a Colab-compatible Jupyter notebook for easy execution and exploration of the model, along with a professional README file detailing the project overview, technical stack, and functionality. This project serves as an excellent example of predictive analytics in consumer electronics.

1. Introduction

In today's rapidly evolving consumer electronics market, understanding the factors that influence product pricing is crucial for both consumers and manufacturers. The Laptop Price Predictor project addresses this need by developing a machine learning model capable of estimating laptop prices based on their technical specifications. This project provides a clear and practical application of regression analysis, a fundamental concept in machine learning, to a real-world problem. The goal is to demonstrate how data-driven insights can be used to predict market values and understand pricing dynamics.

2. Project Objectives

The primary objectives of the Laptop Price Predictor project are:

- To develop a robust Linear Regression model for predicting laptop prices.
- To identify key hardware specifications that significantly impact laptop pricing.
- To implement data preprocessing techniques, including handling categorical variables and feature scaling.
- To provide a clear and well-documented codebase, including a Colab-compatible Jupyter notebook for ease of use and reproducibility.
- To evaluate the model's performance using relevant metrics such as Mean Squared Error (MSE) and R-squared.

3. Methodology

The project utilizes a **Linear Regression** algorithm, a supervised learning model that finds the linear relationship between input features and a continuous output variable (laptop price). The methodology involves several key steps:

- **Data Collection and Preprocessing:** A dataset containing various laptop specifications (e.g., Brand, RAM, Storage, Processor, GPU, Screen Size) and their corresponding prices is collected. Categorical features are converted into numerical representations using One-Hot Encoding, and numerical features are scaled to ensure uniform contribution to the model.
- **Model Training:** The preprocessed data is split into training and testing sets. The Linear Regression model is trained on the training data to learn the optimal weights for each feature.
- **Model Persistence:** The trained model and the feature list are saved using `joblib` to allow for future predictions without retraining.
- **Evaluation:** The model's performance is assessed using metrics such as Mean Squared Error (MSE) to quantify prediction accuracy and R-squared to determine the

proportion of variance in the dependent variable that can be predicted from the independent variables.

4. Features and Functionality

The key features of the Laptop Price Predictor include:

- **Data Preprocessing:** Efficient handling of diverse laptop specifications, including categorical variables like ‘Brand’ through One-Hot Encoding.
- **Predictive Modeling:** Utilizes a Linear Regression model to establish relationships between hardware components and price.
- **Model Persistence:** The trained model (`laptop_price_model.pkl`) and feature set (`features.pkl`) are saved for quick deployment and inference.
- **Evaluation Metrics:** Provides clear evaluation of model performance using MSE and R-squared.
- **Colab-Compatible Notebook (`Project_Notebook.ipynb`):** An interactive Jupyter notebook that allows users to run the entire project, from data loading to model prediction, directly in Google Colab.

5. Contribution of Team Members

This project was a collaborative effort by Srustisri Panda, Puja Rani Mishra, and Kajal Roul. Each member contributed significantly to various aspects of the project:

- **Srustisri Panda (sru-codes):** Focused on data collection, initial data cleaning, and the implementation of the Linear Regression model. Responsible for setting up the project repository and ensuring the `train.py` script functionality.
- **Puja Rani Mishra (mishrapujarani11-web):** Contributed to feature engineering, hyperparameter tuning, and model evaluation. Assisted in creating the project documentation and ensuring the clarity of the README file.
- **Kajal Roul (kajalroul2007-ops):** Primarily responsible for developing the Colab-compatible notebook (`Project_Notebook.ipynb`), ensuring its reproducibility and ease of use. Also contributed to the overall project report and testing of the model.

6. Conclusion

The Laptop Price Predictor successfully demonstrates the application of Linear Regression in predicting consumer electronics prices. The project provides a clear, functional, and well-documented solution for estimating laptop values based on their specifications. This tool can be valuable for consumers making purchasing decisions and for businesses analyzing market trends. The collaborative effort ensured a robust model and comprehensive documentation, making it a strong example of a minor project in predictive analytics.

References

[1] Kaggle. (n.d.). *Laptop Price Prediction Dataset*. Link